```
import java.util.Scanner;
    private static Scanner input = new Scanner(System.in);
    // Global 2D array for hotel rooms
    // 0 = Available, 1 = Occupied
    private static int[][] rooms;

    public static void main(String[] args) {
        System.out.println("================================================");
        System.out.println("            HOTEL ROOM MANAGER");
        System.out.println("================================================");
        System.out.println(" Welcome to the Hotel Room Manager");
        System.out.println(" Manage room occupancy and guest records");
        System.out.println("================================================");
        initialiseRooms();
        runMenu();
    }

    public static void initialiseRooms() {

        rooms = new int[3][]; // 3 floors
        rooms[0] = new int[20]; // Floor 1 has 10 rooms
        rooms[1] = new int[25]; // Floor 2 has 15 rooms
        rooms[2] = new int[15]; // Floor 3 has 12 rooms
        // All rooms are automatically initialised to 0 (available)
    }
    public static void runMenu() {
        int option;
        do {
            option = getOption();

            switch (option) {
                case 1:
                    checkInGuest();
                    break;
                case 2:
                    checkOutGuest();
                    break;
                case 3:
                    showRooms();
                    break;
                case 0:
                    System.out.println("Thank you for using Hotel Room Manager.");
                    break;
                default:
                    System.out.println("Invalid option. Please try again.");
            }

        } while (option != 0);
    }

    public static int getOption() {
        System.out.println();
        System.out.println("+--------------------------------------+");
        System.out.println("|              MAIN MENU               |");
        System.out.println("+--------------------------------------+");
        System.out.println("| 1) Check in guest                    |");
        System.out.println("| 2) Check out guest                   |");
        System.out.println("| 3) Show room availability            |");
        System.out.println("| 0) Quit                              |");
        System.out.println("+--------------------------------------+");
        System.out.print("Please select an option: ");
        return input.nextInt();
    }

```

```
65
66          public static void checkInGuest() {
67
68              System.out.print("Enter floor number: ");
69              int floorNumber = input.nextInt();
70
71              System.out.print("Enter room number: ");
72              int roomNumber = input.nextInt();
73
74              int floorIndex = floorNumber - 1;
75              int roomIndex = roomNumber - 1;
76
77              if (rooms[floorIndex][roomIndex] == 0) {
78                  rooms[floorIndex][roomIndex] = 1;
79                  System.out.println("Check-in successful.");
80              } else {
81                  System.out.println("Room already occupied.");
82              }
83          }
84
85          public static void checkOutGuest() {
86
87              System.out.print("Enter floor number: ");
88              int floorNumber = input.nextInt();
89
90              System.out.print("Enter room number: ");
91              int roomNumber = input.nextInt();
92
93              int floorIndex = floorNumber - 1;
94              int roomIndex = roomNumber - 1;
95
96              if (rooms[floorIndex][roomIndex] == 1) {
97                  rooms[floorIndex][roomIndex] = 0;
98                  System.out.println("Check-out successful.");
99              } else {
100                 System.out.println("Room already available.");
101             }
102         }
103
104         public static void showRooms() {
105
106             System.out.println();
107             System.out.println("================================================");
108             System.out.println("HOTEL ROOM STATUS");
109             System.out.println("================================================");
110             System.out.println("LEGEND: [O] = Available, [X] = Occupied");
111             System.out.println("------------------------------------------------");
112
113             for (int floor = 0; floor < rooms.length; floor++) {
114
115                 System.out.print("Floor " + (floor + 1) + " ");

116
117                 for (int room = 0; room < rooms[floor].length; room++) {
118                     if (rooms[floor][room] == 0) {
119                         System.out.print("[O]");
120                     } else {
121                         System.out.print("[X]");
122                     }
123                 }
124                 System.out.println();
125             }
126
127             System.out.println("================================================");
128     }
129 }
```